
SC/DPS Software Ownership

A clear ownership plan for slow controls, DAQ interfaces, Computing support, and protection boundaries

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DUNE
DEEP UNDERGROUND
NEUTRINO EXPERIMENT

 **Fermilab**

The Whole Proposal

ONE QUESTION

Who owns the software logic that represents detector hardware to operators, DAQ, Computing, and protection systems?

- ▶ Start with ownership, then choose repository names.
- ▶ Keep device-control logic close to Slow Controls when SC owns the bridge.
- ▶ Keep DAQ and Computing interfaces explicit and testable.

Slow Controls

Owns bridge logic, OPC-UA representation, operator alarms, history, and SC-facing command semantics.

DAQ and Computing

DAQ owns DAQ clients and run-control behavior. Computing owns platform support, CI, artifacts, and deployment infrastructure.

DPS / facilities

Owns credited protection and hard interlock logic. SC may display it, but does not become the safety owner.

Online Sources Only

Slow-controls documents

- [Common SC project](#)
- [PDS SC](#)
- [BDE SC](#)
- [HVS SC](#)
- [TDE SC](#)
- [DAQ SC](#)

Installation and protection

- [DPS](#)
- [VD rack assignments](#)
- [HD rack assignments](#)
- [DAQ rack build](#)
- [HWDB exports](#)

Software spaces

- github.com/DUNE-DAQ
- github.com/DUNE
- [DUNE Computing basics](#)
- [Offline Computing CDR](#)

Rule

A production repository should point to the online source it implements. Local notes and prototypes can motivate work, but they are not the authority.

Systems We Need To Cover

Detector electronics

PDS / DAPHNE
TPC / BDE / WIB
TDE / timing

Board services, readiness,
status, commands.

Power and racks

PDU outlets
Rack sensors
IPMI / BMC
Installed identity, outlet maps,
host health.

Detector services

HVS
Calibration
LCM / GIZMo
Permitted operation,
readback, safe ranges.

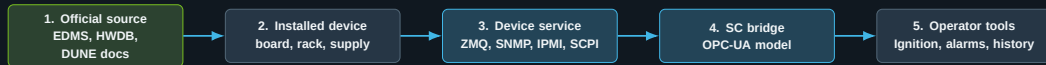
Protection and platform

DPS / facilities
DAQ
Computing
Safety authority, DAQ
consumers, deployment
support.

Interpretation

These are not all the same kind of software. Some are device-control paths, some are consumers, and some are safety or platform responsibilities.

Where Software Goes

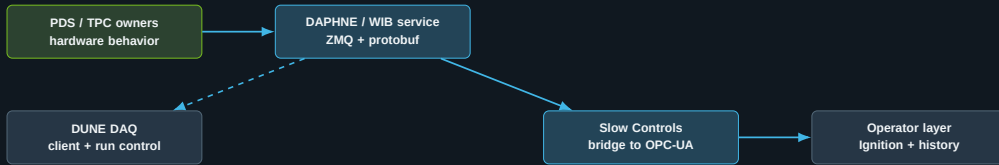


**Subsystem owners define
hardware behavior.**

**Slow Controls owns the
bridge and operator
representation.**

**DAQ and Computing
consume or support stable
contracts.**

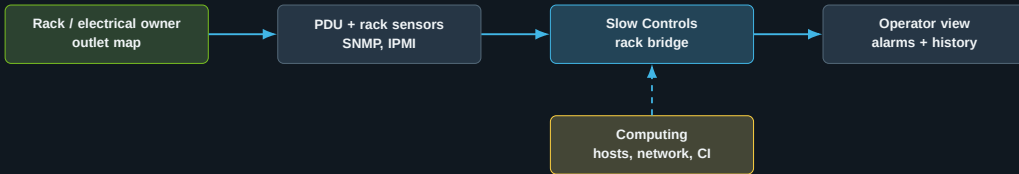
DAPHNE And WIB Boundary



Plain-language boundary

DAQ owns the DAQ client. Slow Controls owns the SC bridge and operator semantics. The shared service contract must be reviewed by both sides.

PDU And Rack Boundary



Plain-language boundary

Slow Controls can monitor and command rack services through approved bridges. Electrical, rack, facilities, and Computing owners keep their separate authority for installation facts and platform operation.

Protection Is A Separate Lane

Observe

SC displays protection state, alarms on it, and archives it for operators.

This is monitoring.

Request

SC may request permitted actions through approved command paths.

This is controlled operation.

Protect

DPS, facilities, and approved interlocks own credited protection.

This must survive ordinary software failures.

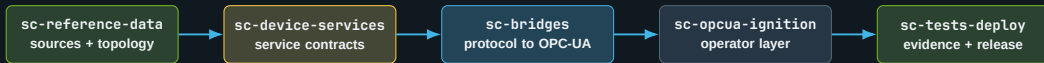
Hard rule

Do not credit a bridge, HMI, GitHub workflow, or normal service process as protection unless the designated protection owner explicitly approves that claim.

Ownership Matrix

System	Slow Controls owns	Other required owners
PDS / DAPHNE	SC bridge and operator state OPC-UA mapping, quality, alarms, history, emulator tests.	PDS + DAQ Board behavior and DAQ client compatibility.
TPC / BDE / WIB	BDE/WIB SC bridge Readiness state, bridge behavior, operator-facing commands.	TPC/BDE + DAQ WIB behavior and run-control integration.
PDU / rack services	Rack monitoring bridge SNMP/IPMI mapping, state-data rules, alarm exposure.	Electrical + Computing Outlet map, rack installation, hosts, network.
HVS / calibration	Operator representation OPC-UA methods, readback quality, alarm/history model.	Subsystem + safety owner Safe ranges, permits, approved modes.
DPS / facilities	Visibility only unless approved Display, alarm, archive, and documented requests.	DPS / facilities Credited protection and hard interlocks.

Repo Families Follow The Layers



Naming principle

The names are examples. The important point is that each repo family has a clear responsibility, owner set, review path, and test obligation.

What Git Must Prove

Source

Which online EDMS, HWDB, DUNE, or GitHub source is implemented?

Owner

Who reviews hardware behavior, SC behavior, DAQ impact, safety impact, and deployment impact?

Contract

Which protocol/schema version is used, and which consumers are compatible?

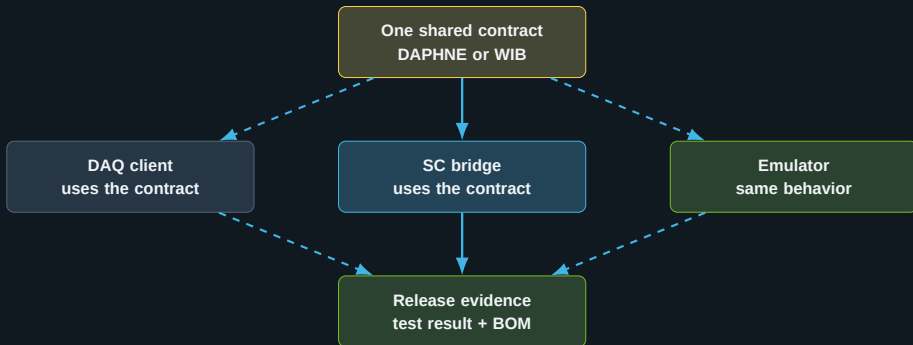
Safety

What fails closed, what is only monitoring, and who signs protection assumptions?

Minimum release evidence

A tagged release should include a tested producer, bridge, emulator scenario, DAQ compatibility result when relevant, deployment manifest, and rollback note.

First Vertical Slice



Why this proves the model

One contract tested by a DAQ client, an SC bridge, and an emulator demonstrates the ownership boundary without requiring all repositories to be finalized first.

Decisions For Thursday

- ▶ Approve the ownership rule before debating final repository names.
- ▶ Confirm that SC-owned bridges govern DAPHNE/WIB slow-control behavior.
- ▶ Choose the governed Git namespace or coordinated namespace policy.
- ▶ Name first reviewers for SC, DAQ, Computing, subsystem, and DPS/facilities.
- ▶ Select the first vertical slice: DAPHNE or WIB.

Meeting target

Agree on boundaries and first owners. Do not require perfect repo names to start.

Technical target

Run one DAQ client and one SC bridge against the same emulator scenario.